

Do the following tasks in your groups (very important to delegate well)

Write code in a Python notebook for the questions provided in Moodle.
(Submit your .ipynb as an .html file)

Do not forget to provide an acceptable MECE table

For the given data write and run regex in your jupyter notebook to answer the following questions. Mention your answers clearly. (60%)

1. How many records have a date that is expressed without using alphabets?
2. How many records have a word starting with the letter “w”?
3. How many records make a word that starts with an alphabet and is not a URL?
4. How many tweets contain one of these emojis :, :D, ;), :P?
5. How many records contain a decimal number?
6. What is the total number of ip addresses across all the records?
7. How many records have a new line?
8. What is the total number of hashtags across all these tweets?
9. What is the code to substitute all non-alphanumeric characters with a new line?
10. What is the total number of URLs across all tweets?

B. Perform stemming and lemmatization (40%)

1. Use porter stemmer to run stemming. Count the number of unique words/tokens before and after stemming
2. Perform lemmatization using NLTK lemmatizer. Count the number of unique words/tokens before and after lemmatization
3. Compare the change in word frequencies from stemming and lemmatization. Which are the top 10 words after stemming/lemmatization
4. What is the change in word frequencies if normalization is done after stop word removal.